Measurement Science
and Technology

PURPOSE-LED
PUBLISHING™

ACCEPTED MANUSCRIPT

Suppressing surface patch potentials on AuPt alloy surfaces for space-based gravitational wave test masses

To cite this article before publication: rui luo *et al* 2026 *Meas. Sci. Technol.* in press https://doi.org/10.1088/1361-6501/aea882

Manuscript version: Accepted Manuscript

Accepted Manuscript is “the version of the article accepted for publication including all changes made as a result of the peer review process, and which may also include the addition to the article by IOP Publishing of a header, an article ID, a cover sheet and/or an ‘Accepted Manuscript’ watermark, but excluding any other editing, typesetting or other changes made by IOP Publishing and/or its licensors”

This Accepted Manuscript is **© 2026 IOP Publishing Ltd. All rights, including for text and data mining, AI training, and similar technologies, are reserved.**.

During the embargo period (the 12 month period from the publication of the Version of Record of this article), the Accepted Manuscript is fully protected by copyright and cannot be reused or reposted elsewhere.

As the Version of Record of this article is going to be / has been published on a subscription basis, this Accepted Manuscript will be available for reuse under a CC BY-NC-ND 4.0 licence after the 12 month embargo period.

After the embargo period, everyone is permitted to use copy and redistribute this article for non-commercial purposes only, provided that they adhere to all the terms of the licence https://creativecommons.org/licences/by-nc-nd/4.0

Although reasonable endeavours have been taken to obtain all necessary permissions from third parties to include their copyrighted content within this article, their full citation and copyright line may not be present in this Accepted Manuscript version. Before using any content from this article, please refer to the Version of Record on IOPscience once published for full citation and copyright details, as permissions may be required. All third party content is fully copyright protected, unless specifically stated otherwise in the figure caption in the Version of Record.

View the article online for updates and enhancements.

This content was downloaded from IP address 115.156.143.209 on 04/10/2026 at 14:10

Suppressing Surface Patch Potentials on AuPt Alloy Surfaces for Space-Based Gravitational Wave Test Masses

Rui Luo, Aoxue Xiao, Yi Yan, Haiyan Zhao, Jinquan Liu, Butian Zhang, Shun Wang, Hang Yin, Pengshun Luo*, Zebing Zhou

National Gravitation Laboratory, MOE Key Laboratory of Fundamental Physical Quantities Measurement, and School of Physics, Huazhong University of Science and Technology, Wuhan 430074, People's Republic of China

E-mail: pluo2009@hust.edu.cn

20 August 2026

Abstract. Patch potentials on conductive surfaces introduce significant electrostatic disturbances in high-precision experiments, including space-based gravitational wave detection. In this work, we systematically investigate the effects of various surface treatment processes—including organic cleaning, acid cleaning, ion sputtering, thermal annealing, gold coating, and atmospheric exposure—on the surface potential distribution of the AuPt alloys that were developed for fabricating the test masses used in space-based gravitational wave detection. Kelvin probe force microscopy characterization reveals that while conventional organic cleaning and thermal annealing offer limited improvements, acid cleaning significantly suppresses potential fluctuations, reducing the root-mean-square values by over a factor of two. Notably, ion sputtering proves to be the most effective technique, consistently yielding highly homogeneous surfaces and producing an approximately fourfold reduction in the root-mean-square values. Auger electron spectroscopy confirms that this enhancement stems from the thorough removal of carbon-, sulfur-, and oxygen-containing surface contaminants. Furthermore, the treated surfaces exhibit robust environmental stability, retaining their homogeneity even after one month of exposure to ambient conditions in a clean environment. These findings provide essential guidance for optimizing surface-treatment strategies in applications such as high-precision inertial sensors.

Keywords: patch potentials, kelvin probe force microscopy, space-based gravitational wave detection

Accepted Manuscript

1. Introduction

An ideal conductor forms an equipotential surface. In principle, no electrostatic force should exist between two such conductors with identical potential. In reality, however, a conductor's surface potential is often inhomogeneous due to crystalline facets, impurities, and surface adsorbates or contaminants. This phenomenon has been known as patch potential for a long time [1–4]. This potential variation can originate from the crystal-orientation-dependent nature of the work functions [5]. Furthermore, the adsorption of atoms or molecules can induce charge transfer, forming local electrical dipoles that alter the local work function [6–8]. Impurities and crystalline defects can also modify the local electronic structure, creating additional potential anomalies [9–12].

The patch potential makes it impossible to fully compensate for the electrostatic forces between two conductors by applying a single voltage between them. Furthermore, as the surface dipoles or potential anomalies fluctuate over time, the patch potential also generates electric-field noise. These effects pose significant challenges for high-precision scientific experiments. For example, extensive investigations have been performed to understand the extra heating rates of the ions by the electric-field noise [13–17]. Patch potentials also play an important role in precision force measurements. In Casimir experiments, the patch electrostatic forces have been considered as a possible source of deviations between experimental measurements and theoretical predictions based on the Drude model [18–22]. Similar electrostatic effects have been investigated in short-range gravity experiments, where uncontrolled surface potentials may introduce additional forces in the test of the gravitational inverse-square law [23–27]. Patch-potential-related effects are also relevant to precision detection of exotic interactions [28–31]. Electrostatic disturbances caused by patch potentials also pose a major challenge for space-based precision experiments. In Gravity Probe B, most of the gyro disturbance effects can be explained by electrostatic patch potentials on both the gyro rotors and their surroundings [32]. Stray electric-field-related noise in gravitational reference sensors has to be mitigated to reduce the charge-related electrostatic forces on free-falling test masses in space-based gravitational wave detection [33–37].

The significant influence of patch potentials on sensitive scientific experiments motivates the development of methods to evaluate and mitigate them. Several techniques exist for measurement. Kelvin probe force microscopy (KPFM), based on atomic force microscopy, offers high spatial resolution imaging of surface potentials [20, 38, 39]. For exceptional sensitivity, torsion balances have achieved record-high measurements [40–43]. Additionally, the effects of patch potentials can be inferred from the heating rates of trapped ions [17].

Various methods have also been applied to reduce patch potentials on conductive surfaces. For instance, ultraviolet (UV) laser pulses have been shown to reduce the heating rate of a single trapped $^{40}Ca^{+}$ ion [15] through patch potential mitigation. Furthermore, ion sputtering effectively reduces surface contamination, thereby minimizing patch potential variations [16, 17]. A combined approach,

Accepted Manuscript

demonstrated by Liu et al., uses UV light followed by Ar^+ ion beam irradiation to greatly eliminate electrostatic patch effects [22]. This combined approach works because UV light removes water and organic contaminants via photosensitized oxidation, while Ar^+ beams effectively sputter away inorganic contaminants. Thermal annealing has also been investigated as a possible method for reducing patch potentials by modifying surface conditions. Garrett et al. showed that among the samples they prepared, the largest patches are found on the sample annealed at 200°C for 3 hours in ambient atmosphere [44].

In this work, we investigate the patch potential on gold-platinum alloys, developed specifically for fabricating low-magnetic test masses for space-based gravitational wave detection [45, 46]. The evolution of the surface potential is studied using KPFM and Auger electron spectroscopy (AES) following different surface processing, including conventional organic-solvent cleaning, acid cleaning, ion sputtering, thermal annealing, and gold coating. The effects of these treatments are quantified through surface potential root-mean-square (RMS) values, and the underlying mechanisms are discussed alongside chemical composition analysis. Based on these findings, we propose an optimized surface processing protocol to effectively suppress patch potential effects on AuPt alloy test masses. Section 2 describes the surface treatment processes and characterization techniques for the samples. Section 3 discusses in detail the influence of different surface treatment processes on the surface potential of the AuPt alloy. Conclusions are presented in Section 4.

2. Experimental details

2.1. Sample preparation and surface treatments

The AuPt alloy, with a nominal composition of 71 at.% Au and 29 at.% Pt, was fabricated by cold plastic deformation. The alloy exhibits a remanent magnetic moment of less than 15 $nA{\cdot}m^2$, as measured by a compound pendulum [46], and is intended for use as test masses for space-based gravitational wave detection [47]. The AuPt alloy was cut into $6 \times 6 \times 2$ mm^3 specimens, with one surface polished to an optical mirror finish. Five samples (S1–S5) were subjected to various surface treatment procedures as summarized in Table 1.

Conventional organic-solvent cleaning: All samples (S1–S5) underwent a conventional organic-solvent cleaning procedure involving sequential 5-min ultrasonic baths in analytical-grade acetone, isopropanol, and ethanol, followed by rinsing with deionized (DI) water to remove residual solvents. They were then dried with high-purity nitrogen and transferred to an ultra-high vacuum chamber for surface potential measurements.

Acid cleaning: Samples S3 and S4 were additionally immersed in an acid solution composed of 30% sulfuric acid and hydrogen peroxide at a 7:1 ratio for 1 minute to remove residual contaminants, and were then rinsed thoroughly with DI water and

Accepted Manuscript

Table 1: Summary of surface treatment procedures for AuPt alloy samples.

Sample	Surface treatment procedures
S1	Conventional organic-solvent cleaning; first ion sputtering; second ion sputtering; thermal annealing at 500 °C; gold coating
S2	Conventional organic-solvent cleaning; thermal annealing at 150 °C, 300 °C, and 500 °C; ion sputtering
S3	Conventional organic-solvent cleaning; acid cleaning; ion sputtering; atmospheric exposure one week and then one month
S4	Conventional organic-solvent cleaning; acid cleaning; ion sputtering
S5	Conventional organic-solvent cleaning; AES characterization before and after ion sputtering in the same vacuum chamber

dried.

Ion sputtering: All samples (S1–S5) underwent ion sputtering at certain stages using an Ar^+ ion source in an argon atmosphere at 5×10^{-5} Torr. The ion acceleration voltage and beam current were set at 2000 V and approximately 10 μA, respectively, with the ion beam incident at 20° relative to the sample surface normal. To ensure uniform treatment, the sample orientation was alternately reversed every 5 minutes for a total sputtering duration of 30 minutes. The high-energy ion beam effectively removes surface contaminants via physical sputtering.

Thermal annealing: Following conventional organic-solvent cleaning, sample S2 was annealed in the preparation chamber of the KPFM under a vacuum of 5×10^{-9} Torr, sequentially at temperatures of 150 °C, 300 °C, and 500 °C. Each annealing step lasted 12 hours, after which the sample was cooled to room temperature at a rate of 5 °C/min and then transferred to the KPFM measurement chamber without exposure to the atmosphere.

Gold coating: A 100-nm-thick gold film was deposited onto sample S1 after ion sputtering. The deposition was carried out by electron beam evaporation at a rate of 0.07 nm/s in the KPFM preparation chamber at room temperature under a base pressure of 2.2×10^{-9} Torr. After the KPFM measurements were completed, the sample was further annealed at 500 °C and then measured again.

Atmospheric exposure: After the initial measurements, sample S3 was further exposed to ambient air in a cleanroom for one week and one month, respectively, to evaluate the stability of its surface potential under ambient conditions.

Ion sputtering, thermal annealing, and gold coating were conducted in the KPFM preparation chamber, which is interconnected with the KPFM measurement chamber through a gate valve. This *in situ* workflow is essential for preserving a clean surface by avoiding air exposure during sample transfer, thereby allowing stringent control over surface contamination.

Accepted Manuscript

2.2. Surface characterization

The surface potential was measured by KPFM in dual-frequency mode under ultra-high vacuum (see Fig. 1). Unlike the conventional dual-pass KPFM method, this mode enables the simultaneous acquisition of topography and surface-potential images in a single scan. In the dual-frequency mode, the cantilever is excited at two distinct eigenmodes. The first eigenmode [$f_1 = 15.02$ (2) kHz] is used for topography imaging via frequency-modulated atomic force microscopy mode (FM-AFM). The second eigenmode [$f_2 = 94.02$ (4) kHz] is dedicated to surface potential measurement based on amplitude-modulation KPFM (AM-KPFM). The voltage applied between the conductive tip and the sample can be expressed as:

$$V_{\mathrm{Tip}} = V_{\mathrm{DC}} + V_{\mathrm{AC}} \sin(2\pi f_2 t), \tag{1}$$

where V_{DC} is the compensation voltage and V_{AC} is the AC modulation voltage amplitude. The applied AC voltage generates an electrostatic force component at the modulation frequency (f_2):

$$F_\omega \propto \frac{\partial C}{\partial z}(V_{\mathrm{DC}} - V_{\mathrm{CPD}})V_{\mathrm{AC}} \sin(2\pi f_2 t), \tag{2}$$

where C is the tip-sample capacitance and V_{CPD} is the local contact potential difference. A lock-in amplifier detects the cantilever response at the modulation frequency f_2, and a feedback loop adjusts V_{DC} until the electrostatic force component is minimized. Under this null condition:

$$V_{\mathrm{DC}} = V_{\mathrm{CPD}}. \tag{3}$$

Scanning the probe across the surface provides a spatial map of the local contact potential difference.

In AM-KPFM mode, the absolute magnitude of the measured patch potentials is known to be underestimated due to spatial convolution effects within the tip–cantilever–sample system [48, 49]. While a quantitative deconvolution requires detailed knowledge of the tip geometry and tip-sample distance–which could not be fully determined in this study–we estimate that this effect reduces the measured RMS values by approximately 46% based on typical probe dimensions and a 10 nm tip–sample separation. The deconvolution is performed using a three-dimensional asymmetric point spread function, which is derived from the local capacitance gradients of the tip apex (sphere-to-plane model) and a 2-dimensional (2D) Gaussian density function modeling the cone (Hudlet model) and the cantilever [48, 49]. To ensure statistical reliability, each sample is characterized across four discrete regions. Particular care is taken to monitor and maintain the spatial resolution of the probe tip, ensuring consistent potential evaluations across different datasets.

The KPFM measurements were performed on a home-built scanning probe microscope, as described in Ref. [29]. Conductive silicon probes (ContE-G) were employed, featuring 5-nm chromium and 25-nm platinum coatings on both sides of the cantilever. The typical tip radius is less than 25 nm, with a nominal spring constant

Accepted Manuscript

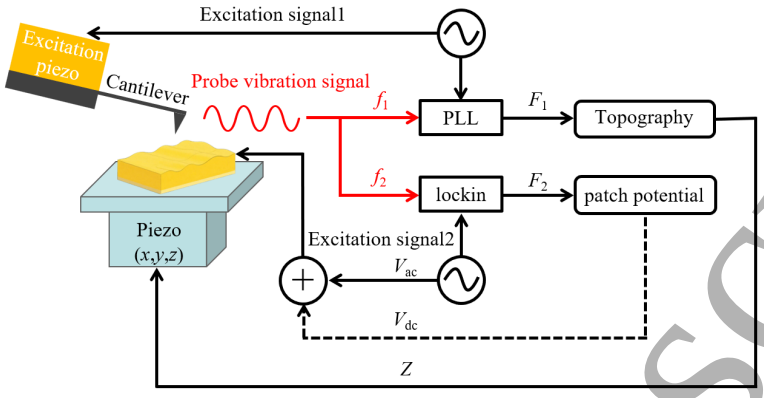


Figure 1: Schematic diagram of dual-frequency KPFM.

of 0.2 N/m. Images were typically acquired at a scan rate of 2 μm/s with a grid size of 256 × 256 pixels. Raw data were analyzed using the Gwyddion software [50]. To highlight the spatial variations in surface potential, the mean value was subtracted from the raw KPFM data for each image. The mean value represents the average CPD, which stems from the work-function difference between the tip and sample, along with any instrumental offsets. The RMS value of the resulting zero-centered data was then calculated to quantify the microscopic surface-potential fluctuations.

The precision of the potential measurement was evaluated by recording a V_{DC} time series while keeping the probe stationary relative to the sample. The standard error of the measured V_{DC} was used to characterize the voltage measurement stability. For a 1 s data acquisition interval, the standard error was calculated to be approximately 2.9 mV and 1.6 mV for modulation voltages of 0.5 V and 0.75 V, respectively.

AES was employed to monitor the compositional evolution of the AuPt alloy surface and assess the impact of ion sputtering for sample S5. By analyzing the kinetic energy of emitted Auger electrons, this technique offers exceptional surface sensitivity with a probing depth of 3-5 nm, permitting a detailed assessment of the chemical nature and relative concentration of surface contaminants.

3. Results and discussions

All samples are first cleaned using conventional organic solvent protocols widely employed in semiconductor manufacturing processes. Figures 2(a)-2(d) present the KPFM measurement results for samples S1 and S2 obtained immediately after the initial organic cleaning step. As illustrated in the topographic images, surface roughness is approximately 3 nm [see Figs. 2(a) and 2(c)]; however, residual mechanical polishing tracks remain clearly discernible. Notably, no significant correlation is observed between the surface potential patterns and the residual polishing tracks. This indicates that the surface potential is nearly independent of local surface curvature, a result consistent with the principles of classical electromagnetics for an ideal conductor. The surface

Accepted Manuscript

potential exhibits significant spatial fluctuation, with RMS values of 29.7 mV and 20.2 mV for samples S1 and S2, respectively. The RMS values typically exceed 15 mV for most measured samples subjected solely to organic cleaning. In the potential images, numerous dark pits exhibiting significant deviations from the mean correlate with bright features in the topographic images. This correlation suggests that local contaminants are the source of these surface potential anomalies, with representative locations indicated by red arrows in the figures.

In the following subsections, we evaluate the effects of various surface treatment protocols following the initial cleaning procedure.

3.1. Acid Cleaning

Acid cleaning following the conventional organic-solvent cleaning effectively reduces surface potential fluctuations, as demonstrated in Figs. 2(f) and 2(h) for samples S3 and S4. The RMS values decrease significantly to 4.8 and 6.4 mV, respectively. RMS values of approximately 2.6 - 7.9 mV are consistently obtained at various positions on both samples. It is widely recognized that acid cleaning effectively removes metal oxides and tenacious hydrocarbons that are difficult to eliminate by conventional organic-solvent cleaning alone, thereby yielding a cleaner and more uniform surface.

3.2. Ion sputtering

To systematically evaluate the effect of ion sputtering, sample S1 is subjected to two consecutive sputtering cycles, with KPFM measurements performed after each step. As illustrated in Fig. 2(b), the organic-solvent cleaned S1 sample initially exhibits a high surface potential RMS value of approximately 29.7 mV. After the first ion sputtering, this value decreases to 4.2 mV [Fig. 3(b)]. Similar enhancements resulting from ion sputtering are consistently observed in other comparative groups, including the acid-cleaned S4 and the AES-measured S5 samples. Notably, as shown in Figs. 3(c) and 3(d), a second ion sputtering cycle on sample S1 yields no further discernible improvement in surface potential. This suggests that a single cycle is sufficient to thoroughly remove surface contaminants. However, while the residual polishing marks are observed to gradually vanish, the surface roughness increases further compared to that after the first sputtering [Fig. 3(c)]. These observations indicate that although ion sputtering effectively reduces surface potential fluctuations, excessive or repeated treatment may increase surface roughness due to the cumulative effect of physical sputtering.

AES analysis is performed on sample S5 to establish a correlation between surface potential fluctuations and chemical composition. The sample is first cleaned with conventional organic-solvent, and then AES spectra are acquired both before and after *in situ* ion sputtering in the same vacuum chamber. As shown in Fig. 4, the AES spectra of the initial surface exhibit prominent signals for carbon (C), sulfur (S), and oxygen (O), indicating the presence of organic contaminants, residual oxides, and potential sulfurized oil residues from mechanical processing. After ion sputtering, these

Accepted Manuscript

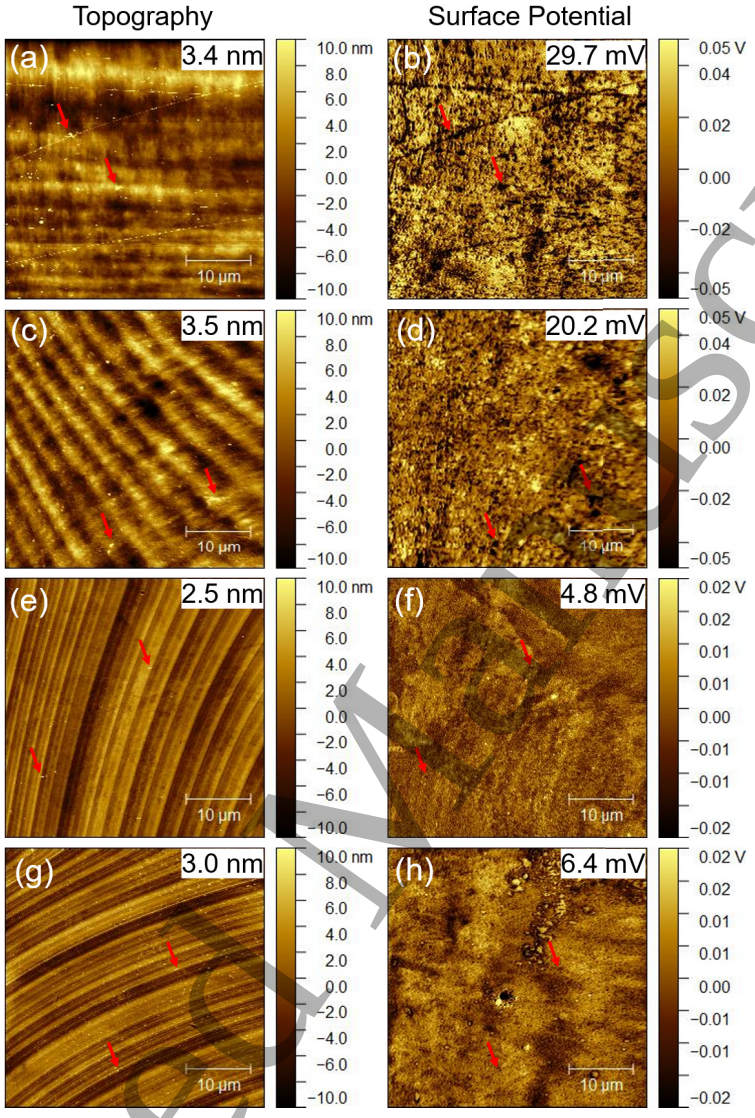


Figure 2: Images of topography (a), (c), (e), (g) and surface potential (b), (d), (f), (h) for different AuPt alloy samples (scan size: 40 µm × 40 µm). The RMS value of topographic roughness and surface potential fluctuations is indicated in the top-right corner of each image. Values in parentheses denote the RMS potential fluctuations measured from three other independent regions, a convention followed in subsequent figure captions. Conventional organic-solvent cleaning: (a), (b) sample S1 (26.5 mV, 25.1 mV, 32.8 mV); (c), (d) sample S2 (15.5 mV, 24.1 mV, 21.3 mV). Additional acid cleaning: (e), (f) sample S3 (6.8 mV, 2.9 mV, 2.6 mV); (g), (h) sample S4 (6.8 mV, 7.9 mV, 7.7 mV).

Accepted Manuscript

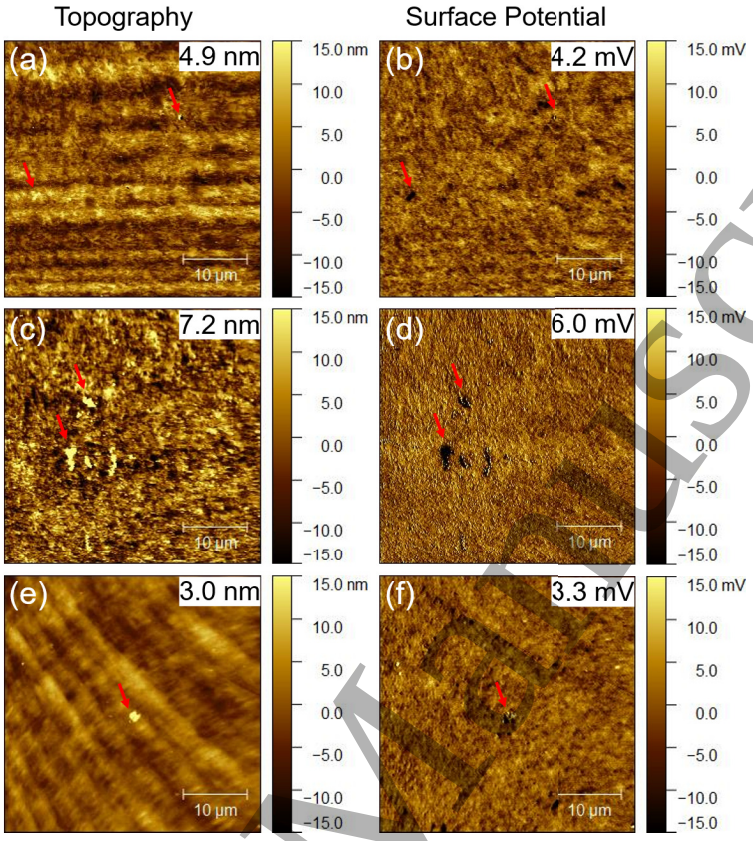


Figure 3: Impact of ion sputtering on the surface topography (left column) and potential (right column): (a), (b) sample S1 (first ion sputtering) (4.5 mV, 5.3 mV, 4.5 mV); (c), (d) S1 (second ion sputtering) (4.8 mV, 7.4 mV, 6.9 mV); and (e), (f) sample S5 (post-AES measurement) (3.0 mV, 4.5 mV, 3.7 mV)

.

contaminant peaks are substantially suppressed, while the characteristic peaks of Pt and Au are enhanced, confirming the effective removal of these surface contaminants. KPFM measurements are subsequently performed on the ion-sputtered S5 sample following AES characterization (The sample is exposed to ambient air during transport between the two measurements). The surface potential RMS value is measured to be 3.3 mV [Fig. 3(f)], consistent with the result obtained for sample S1 after ion sputtering.

These consistent observations across multiple samples indicate that, regardless of different pre-treatments, ion sputtering effectively suppresses surface potential fluctuations. AES measurements show that this method efficiently removes carbon-, sulfur-, and oxygen-containing surface contaminants, thereby improving overall surface cleanliness.

3.3. Thermal Annealing

Thermal annealing is employed to eliminate lattice defects and desorb surface contaminants via thermally activated processes. The effects of thermal annealing are

Accepted Manuscript

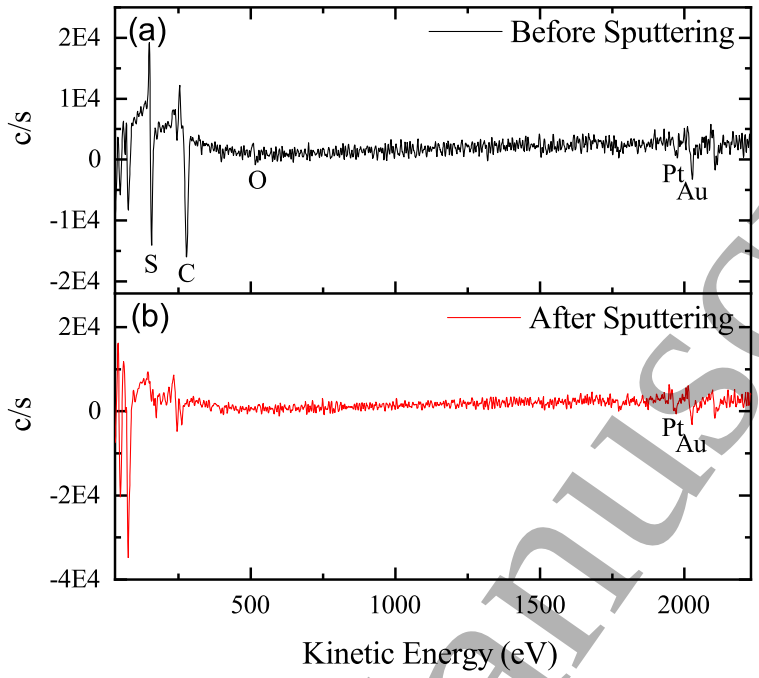


Figure 4: AES spectra of sample S5 before (a) and after (b) ion sputtering: The AES spectroscopy of the initial surface showing prominent signals of carbon (C), sulfur (S), and Oxygen (O); The results recorded after ion sputtering showing the removal of surface contaminants and the enhancement of characteristic Au and Pt peaks.

evaluated by monitoring the evolution of surface potential RMS values across sequential thermal stages at 150 °C, 300 °C, and 500 °C for sample S2. The RMS values are measured to be 22.1 mV, 17.0 mV, and 23.1 mV for the three annealing temperatures [Figs. 5(b), 5(d), and 5(f)], respectively, compared to 20.2 mV for the non-annealed sample [Fig. 2(d)]. No significant improvements are observed in either the surface potential distribution or the surface structures [Figs. 5(a), 5(c), and 5(e)], implying that the thermal energy is insufficient to desorb the surface contaminants, which are primarily composed of organic compounds and oxides, as evidenced by the AES measurements. In contrast, applying ion sputtering to the 500 °C-annealed counterpart results in a substantial reduction of the RMS value to 7.1 mV. The ion sputtering also slightly increases the surface roughness. These findings reaffirm that ion sputtering is highly effective in removing surface contaminants and reducing surface potential fluctuations.

3.4. Gold Coating

Gold is deposited on sample S1 after the sequential treatment of ion sputtering and annealing at 500 °C. The surface potential measurements of the gold-coated sample reveal that the gold layer has a negligible impact on surface potential homogeneity. The RMS values of sample S1 are 6.0 mV after the second ion sputtering, and 4.6 mV with further annealing at 500 °C. Following the gold coating, the RMS value changes slightly

Accepted Manuscript

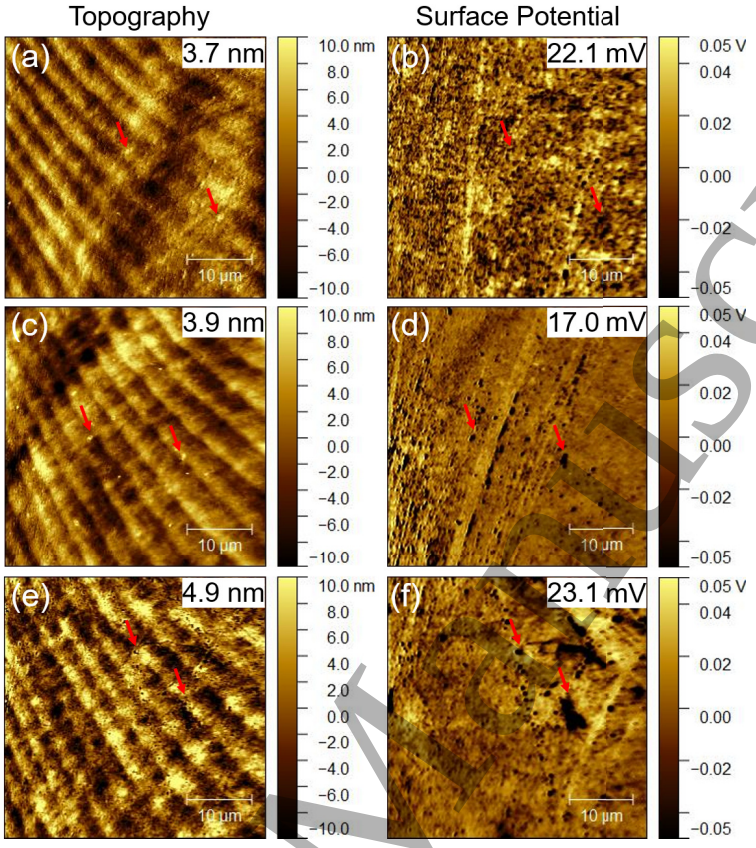


Figure 5: Surface evolution of sample S2 with incremental thermal annealing temperature. Topographic images and potential images are presented in the left and right columns, respectively. Thermal annealing at (a), (b) 150 °C (25.6 mV, 32.7 mV, 27.2 mV); (c), (d) 300 °C (20.8 mV, 19.1 mV, 18.2 mV); and (e), (f) 500 °C (20.6 mV, 21.2 mV, 20.0 mV)

.

to 6.5 mV [Fig. 6(b)], falling within the low-potential-fluctuation range achievable in this study. It should be noted that the RMS values may vary slightly from region to region on the same sample. Given that test masses generally require a gold coating, these findings demonstrate that the deposition of a protective gold film does not undermine the homogeneity of the surface potential.

3.5. Temporal Evolution in Air

To evaluate the stability of the cleaned surface under ambient exposure, a long-term exposure test is conducted on sample S3. The sample is prepared through sequential organic and acid cleaning, followed by ion sputtering to achieve a pristine surface [Fig. 7(a) and 7(b)]. As shown in Fig. 7(b), the initial surface potential RMS is as low as 3.3 mV. Subsequently, the sample is stored in a clean room under ambient conditions. The surface potentials are then re-measured after intervals of one week and one month. As shown in Figs. 7(d) and 7(f), the RMS values after one week and one month

Accepted Manuscript

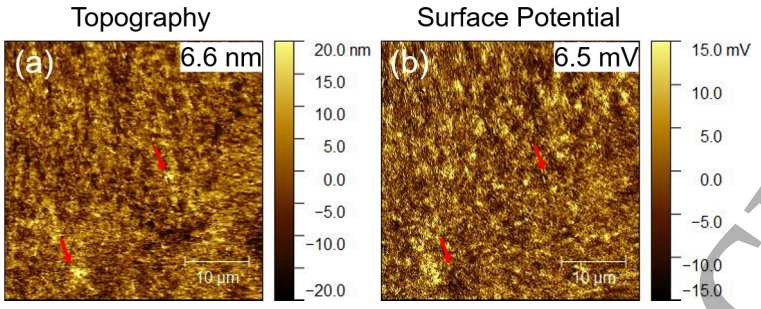


Figure 6: (a) Topographic image and (b) surface potential image of sample S1 taken after 100 nm gold coating (7.7 mV, 6.2 mV, 5.7 mV)

.

of exposure are 3.5 mV and 5.6 mV, respectively. Similar to the surface potential, the surface morphology shows slight variations, with the topographic RMS roughness measured at 5.8 nm after one week and remaining below 9 nm even after a full month of exposure [Figs. 7(a), 7(c), and 7(e)]. These results imply that when stored in a clean environment, such as a cleanroom, the surface of the AuPt alloy can be maintained for a relatively long period. This stability is partly attributed to the intrinsic chemical inertness of the constituent noble metals (Au and Pt), which significantly enhances the alloy's resistance to atmospheric oxidation and adsorption. As demonstrated in previous subsections, the primary sources of surface potential variation are organic and oxide contamination, both of which are effectively mitigated for noble metals in a clean environment. In a cleanroom environment, water and small gas molecules are expected to be the primary adsorbates; these are likely to form a relatively uniform surface layer that does not significantly increase the RMS surface-potential fluctuation.

3.6. Spatial correlation analysis

While the RMS surface-potential fluctuation characterizes the amplitude of patch potentials, a comprehensive evaluation of their electrostatic impact requires quantifying their spatial correlation length scales. To this end, we also perform a spatial autocorrelation analysis following the methodologies in Refs. [19, 49, 51]. The 2D autocorrelation functions, $C(r) = \langle V_i(r)V(0)\rangle$, are calculated from the mean-subtracted KPFM potential images of two typical measurements: the organically cleaned surface (non-sputtered surface) [Fig. 2(b)] and the ion-sputtered surface [Fig. 3(b)]. Figure 8 displays the radial profile of $C(r)$.

To perform a quantitative assessment, we fit the experimental curves using the correlation functions of Model 1 [19, 51]:

$$C(r) = 2V_{rms}^2 \frac{k_{max}J_1(k_{max}r) - k_{min}J_1(k_{min}r)}{(k_{max}^2 - k_{min}^2)r} \tag{4}$$

Accepted Manuscript

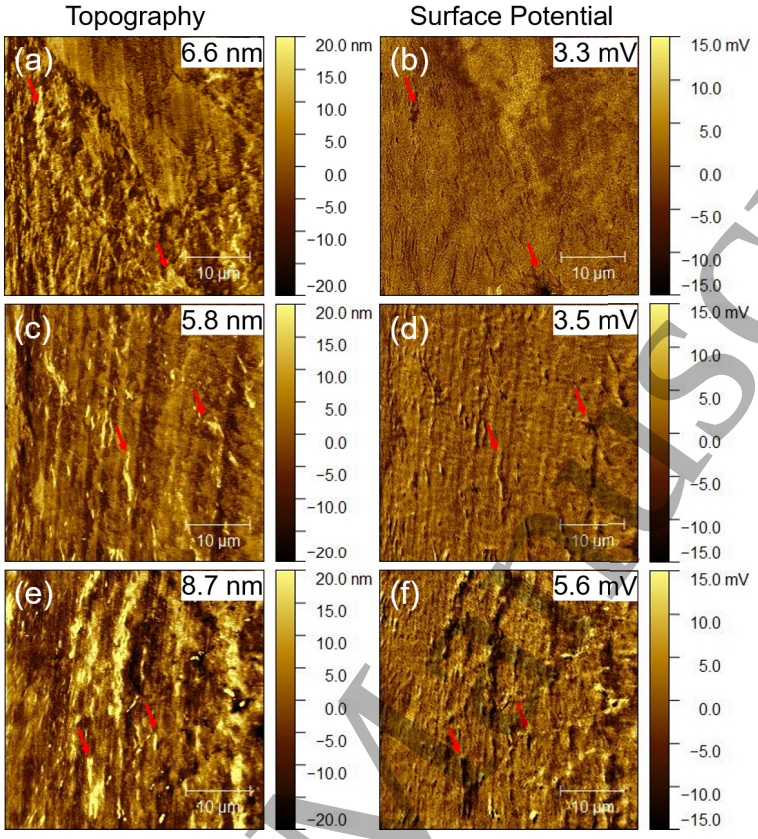


Figure 7: Time-dependent evolution of surface topography (a, c, e) and potential (b, d, f) of sample S3 through ambient exposure: (a), (b) the initial state after ion sputtering (2.7 mV, 2.8 mV, 3.4 mV); (c), (d) one week exposure in air (5.1 mV, 4.4 mV, 3.9 mV); and (e), (f) one month exposure in air (7.2 mV, 5.5 mV, 9.2 mV)

.

and Model 2 [49]:

$$C(r) = V_{rms}^2 \left(1 - \frac{r(4 - r/\lambda)}{\lambda\pi}\right). \tag{5}$$

Fitting the data to Model 1 yields $V_{\rm rms} = 26.8$ mV, $k_{max} = 5.1$ μm^{-1} and $k_{min} = 0$ for the un-sputtered surface, and $V_{\rm rms} = 3.7$ mV, $k_{max} = 4.1$ μm^{-1} and $k_{min} = 0$ for the ion-sputtered surface. For Model 2, the fit yields $V_{\rm rms} = 29.5$ mV and $\lambda = 0.86$ µm for the un-sputtered surface, and $V_{\rm rms} = 4.2$ mV, $\lambda = 0.90$ µm for the ion-sputtered surface. These fits show a reduction in V_{rms} consistent with previous analyses. However, the overall quality of the fits suggests that a more sophisticated model is required to fully describe the patch potential distribution.

To provide an order-of-magnitude estimate of the patch electrostatic force, we consider a parallel-plate configuration with an electrode area of 494.06 mm^2 and a separation of 4.131 mm, which are typical dimensions for the TianQin mission [52]. Using the fitting parameters from Model 2, we estimate the static electrostatic force to be $\sim 1.1 \times 10^{-20}$ N for the ion-sputtered surface. This force could be 3.4 times

Accepted Manuscript

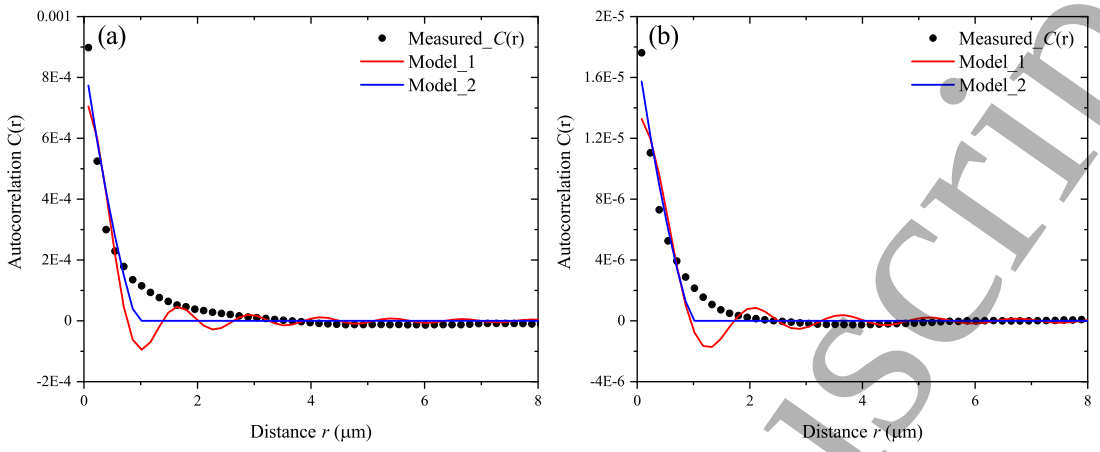


Figure 8: Spatial correlation analysis of patch potentials on (a) the conventional organic-solvent-cleaned surface [see Fig. 2(b)] and (b) the ion-sputtered surface [see Fig. 3(b)]. The red and blue curves represent fits to Model 1 and Model 2, respectively.

larger when accounting for the underestimation of the potential inherent in AM-KPFM measurements. However, the potential distribution cannot be assumed to remain uniform on larger scales—specifically at the millimeter scale—which is critical given the centimeter-scale dimensions of the test mass. A comprehensive, mission-level evaluation requires large-area potential mapping of the full-scale test mass, spatiotemporal correlation analyses, and detailed electrostatic force modeling. These objectives will be pursued in future work using a custom-built KPFM setup.

4. Conclusions

In summary, this study systematically evaluated various surface treatment protocols for AuPt alloys to suppress surface potential fluctuations (patch effects), a critical requirement for high-precision spaceborne gravitational wave detection. We found that acid cleaning is more effective than conventional organic-solvent cleaning in improving surface potential homogeneity, as it more efficiently removes tenacious oxides and hydrocarbons. Thermal annealing (up to 500 °C) proved insufficient for desorbing the specific contaminants present on AuPt surfaces, yielding no discernible improvement in surface potential homogeneity. In contrast, ion sputtering was identified as the most effective technique, consistently reducing surface potential RMS values to below 5 mV. AES measurements confirmed that this improvement stems from the thorough removal of carbon-, sulfur-, and oxygen-containing contaminants. However, excessive or repeated sputtering does not further improve potential homogeneity but instead increases topographic roughness. Notably, the ion-sputtered surfaces exhibit remarkable long-term environmental stability, maintaining surface homogeneity even after one month of ambient exposure in a cleanroom. Based on these findings, we propose a cleaning protocol starting with chemical pre-cleaning (organic solvents and acid

Accepted Manuscript

solutions) followed by a controlled single-cycle ion sputtering. This procedure effectively suppresses surface potential fluctuations and provides a robust surface treatment solution for test masses and other critical components in high-precision space missions. Future work will focus on large-area potential mapping of the full-scale test mass and quantifying its contribution to acceleration noise.

Data availability statement

The data are available from the authors upon reasonable request.

Acknowledgements

This work was supported by the National Key R&D Program of China (Grant Nos. 2022YFC2204100 and 2021YFC2202300), the National Natural Science Foundation of China(NSFC)(Grant No. 12475052), and the Knowledge Innovation Program of Wuhan-Basic Research (Grant No. 2023010201010042).

References

[1] M. J. Morant and H. House, The Work Function and Patch Field of an Irregular Metal Surface, Proc. Phys. Soc. B **69**, 14 (1956).

[2] R. D. Young and H. E. Clark, Effect of Surface Patch Fields on Field-Emission Work-Function Determinations, Phys. Rev. Lett. **17**, 351 (1966).

[3] N. A. Burnham, R. J. Colton, and H. M. Pollock, Work-function anisotropies as an origin of long-range surface forces, Phys. Rev. Lett. **69**, 144 (1992).

[4] C. C. Speake, Forces and force gradients due to patch fields and contact-potential differences, Class. Quantum Grav. **13**, A291 (1996).

[5] G. N. Derry, M. E. Kern, and E. H. Worth, Recommended values of clean metal surface work functions, J. Vac. Sci. Technol. A **33**, 060801 (2015).

[6] V. De Renzi *et al.*, Metal Work-Function Changes Induced by Organic Adsorbates: A Combined Experimental and Theoretical Study, Phys. Rev. Lett. **95**, 046804 (2005).

[7] E. Kim *et al.*, Electric-field noise from carbon-adatom diffusion on a Au(110) surface: First-principles calculations and experiments, Phys. Rev. A **95**, 033407 (2017).

[8] H. Jooya *et al.*, Mechanisms for carbon adsorption on Au(110)-(2 × 1): A work function analysis, Surf. Sci. **677**, 232 (2018).

[9] S. Prakash, Theory of charge perturbation due to impurities in transition metals, J. Phys. F: Met. Phys. **8**, 2497 (1978).

[10] F. Flores, N. March, Y. Ohmura, and A. Stoneham, Asymptotic displaced charge round impurities in metal crystals with and without surfaces, J. Phys. Chem. Solids **40**, 531 (1979).

[11] B. Pal, S. Rattan, J. Singh, and S. Prakash, Electric field gradient for simple metal impurities in Ag and Au, Physica B+C **141**, 203 (1986).

[12] S. Prakash, S. K. Rattan, and J. Singh, Theory of the electric-field gradient in dilute alloys, Phys. Rev. B **48**, 6927 (1993).

[13] N. Daniilidis *et al.*, Fabrication and heating rate study of microscopic surface electrode ion traps, New J. Phys. **13**, 013032 (2011).

[14] M. Brownnutt, M. Kumph, P. Rabl, and R. Blatt, Ion-trap measurements of electric-field noise near surfaces, Rev. Mod. Phys. **87**, 1419 (2015).

Accepted Manuscript

[15] D. T. C. Allcock *et al.*, Reduction of heating rate in a microfabricated ion trap by pulsed-laser cleaning, New J. Phys. **13**, 123023 (2011).
[16] D. A. Hite *et al.*, 100-Fold Reduction of Electric-Field Noise in an Ion Trap Cleaned with *In Situ* Argon-Ion-Beam Bombardment, Phys. Rev. Lett. **109**, 103001 (2012).
[17] D. A. Hite, K. S. McKay, and D. P. Pappas, Surface science motivated by heating of trapped ions from the quantum ground state, New J. Phys. **23**, 103028 (2021).
[18] R. O. Behunin, Y. Zeng, D. A. R. Dalvit, and S. Reynaud, Electrostatic patch effects in Casimir-force experiments performed in the sphere-plane geometry, Phys. Rev. A **86**, 052509 (2012).
[19] R. O. Behunin, F. Intravaia, D. A. R. Dalvit, P. A. M. Neto, and S. Reynaud, Modeling electrostatic patch effects in Casimir force measurements, Phys. Rev. A **85**, 012504 (2012).
[20] R. O. Behunin *et al.*, Kelvin probe force microscopy of metallic surfaces used in Casimir force measurements, Phys. Rev. A **90**, 062115 (2014).
[21] J. Xu, G. L. Klimchitskaya, V. M. Mostepanenko, and U. Mohideen, Reducing detrimental electrostatic effects in Casimir-force measurements and Casimir-force-based microdevices, Phys. Rev. A **97**, 032501 (2018).
[22] M. Liu, R. Schafer, J. Xu, and U. Mohideen, Elimination of electrostatic forces in precision Casimir force measurements using UV and Argon ion radiation, Mod. Phys. Lett. A **35**, 2040001 (2020).
[23] L.-C. Tu, S.-G. Guan, J. Luo, C.-G. Shao, and L.-X. Liu, Null test of newtonian inverse-square law at submillimeter range with a dual-modulation torsion pendulum, Phys. Rev. Lett. **98**, 201101 (2007).
[24] D. J. Kapner *et al.*, Tests of the gravitational inverse-square law below the dark-energy length scale, Phys. Rev. Lett. **98**, 021101 (2007).
[25] W.-H. Tan *et al.*, New test of the gravitational inverse-square law at the submillimeter range with dual modulation and compensation, Phys. Rev. Lett. **116**, 131101 (2016).
[26] W.-H. Tan *et al.*, Improvement for testing the gravitational inverse-square law at the submillimeter range, Phys. Rev. Lett. **124**, 051301 (2020).
[27] J. G. Lee, E. G. Adelberger, T. S. Cook, S. M. Fleischer, and B. R. Heckel, New test of the gravitational $1/r^2$ law at separations down to 52 μm, Phys. Rev. Lett. **124**, 101101 (2020).
[28] Y.-J. Chen *et al.*, Stronger limits on hypothetical yukawa interactions in the 30–8000 nm range, Phys. Rev. Lett. **116**, 221102 (2016).
[29] J. Wang *et al.*, Test of non-newtonian gravitational forces at micrometer range with two-dimensional force mapping, Phys. Rev. D **94**, 122005 (2016).
[30] J. Ding *et al.*, Constraints on the velocity and spin dependent exotic interaction at the micrometer range, Phys. Rev. Lett. **124**, 161801 (2020).
[31] S. Li, W. Zhang, R. Luo, J. Liu, and P. Luo, Improved limits on the spin- and velocity-dependent exotic interaction in the micrometer range, Phys. Rev. Lett. **134**, 251601 (2025).
[32] S. Buchman and J. P. Turneaure, The effects of patch-potentials on the gravity probe B gyroscopes, Rev. Sci. Instrum. **82**, 074502 (2011).
[33] N. A. Robertson *et al.*, Kelvin probe measurements: Investigations of the patch effect with applications to ST-7 and LISA, Class. Quantum Grav. **23**, 2665 (2006).
[34] F. Antonucci *et al.*, Interaction between Stray Electrostatic Fields and a Charged Free-Falling Test Mass, Phys. Rev. Lett. **108**, 181101 (2012).
[35] M. Armano *et al.*, Charge-Induced Force Noise on Free-Falling Test Masses: Results from LISA Pathfinder, Phys. Rev. Lett. **118**, 171101 (2017).
[36] C. Zanoni and D. Bortoluzzi, Estimation of the Electrostatic Effects in the LISA-Pathfinder Critical Test Mass Dynamics via the Method of Moments, IEEE/ASME Trans. Mechatron. **26**, 3022 (2021).
[37] S. Vitale, V. Ferroni, L. Sala, and W. J. Weber, Estimate of force noise from electrostatic patch potentials in LISA Pathfinder, Class. Quantum Grav. **41**, 195009 (2024).
[38] Y. Sugawara *et al.*, High potential sensitivity in heterodyne amplitude-modulation Kelvin probe force microscopy, Appl. Phys. Lett. **100**, 223104 (2012).

Accepted Manuscript

[39] Z. M. Ma, L. Kou, Y. Naitoh, Y. J. Li, and Y. Sugawara, The stray capacitance effect in Kelvin probe force microscopy using FM, AM and heterodyne AM modes, Nanotechnology **24**, 225701 (2013).

[40] S. E. Pollack, S. Schlamminger, and J. H. Gundlach, Temporal Extent of Surface Potentials between Closely Spaced Metals, Phys. Rev. Lett. **101**, 071101 (2008).

[41] H. Yin *et al.*, Measurements of temporal and spatial variation of surface potential using a torsion pendulum and a scanning conducting probe, Phys. Rev. D **90**, 122001 (2014).

[42] K. Li *et al.*, Precision improvement of patch potential measurement in a scanning probe equipped torsion pendulum, Rev. Sci. Instrum. **93**, 065110 (2022).

[43] C. Song *et al.*, A high precision surface potential imaging torsion pendulum facility to investigate physical mechanism of patch effect, Rev. Sci. Instrum. **94**, 024501 (2023).

[44] J. L. Garrett, D. Somers, and J. N. Munday, The effect of patch potentials in casimir force measurements determined by heterodyne kelvin probe force microscopy, J. Phys. Condens. Matter **27**, 214012 (2015).

[45] A. Lou *et al.*, Theoretical calculations and experimental measurements on the two-component Au-Pt alloys with ultralow magnetic susceptibility, Phys. Rev. Appl. **19**, 034080 (2023).

[46] Y. Yu *et al.*, Measuring remanent magnetic moment of test masses by a compound pendulum in a magnetic shielding room, Rev. Sci. Instrum. **96**, 024501 (2025).

[47] Y. Yan *et al.*, Effects of cold plastic deformation on microstructure and magnetic susceptibility of Au-Pt alloys, J. Alloy. Compd. **1007**, 176442 (2024).

[48] A. Liscio *et al.*, Quantitative measurement of the local surface potential of π-conjugated nanostructures: A kelvin probe force microscopy study, Adv. Funct. Mater. **16**, 1407 (2006).

[49] K. Shi, P. Luo, J. Liu, H. Yin, and Z. Zhou, Dependence of the electrostatic patch force evaluation on the lateral resolution of kelvin probe force microscopy, Phys. Rev. D **110**, 122007 (2024).

[50] D. Nečas and P. Klapetek, Gwyddion: an open-source software for spm data analysis, Cent. Eur. J. Phys. **10**, 181 (2012).

[51] C. C. Speake and C. Trenkel, Forces between conducting surfaces due to spatial variations of surface potential, Phys. Rev. Lett. **90**, 160403 (2003).

[52] J. Luo *et al.*, TianQin: A space-borne gravitational wave detector, Classical and Quantum Gravity **33**, 35010 (2016).

Accepted Manuscript